Aerodynamic Analysis of a 1:35 F-35-Style 3D-Printed RC Airframe

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Abstract

This paper estimates the aerodynamic performance of a 446 mm long, 305 mm span, 3D-printed PLA airframe with F-35 proportions, and answers one design question: how much thrust must a booster mounted at the tail supply so that the wings alone carry the aircraft's weight in steady, level flight. Two-dimensional section data for the NACA 0006 wing (and NACA 0004 tail) were computed with XFOIL 6.99 at chord Reynolds numbers from 40,000 to 300,000, in both free-transition and tripped (rough-print) conditions. These were corrected to three dimensions with the Helmbold/DATCOM lift-slope relation and combined with a Raymer component drag build-up.

At an estimated all-up mass of 478 g with a 50 mm electric ducted fan (EDF) in the tail, the airframe stalls at about **40 mph (18 m/s)**, reaches its best lift-to-drag ratio of **7–9** near 55–60 mph, and needs a **minimum sustained thrust of 52–67 gram-force (0.51–0.66 N)**, a thrust-to-weight ratio of only 0.11–0.14. The thrust needed to *get* there is much larger: a booster of **450–770 gf static thrust** is required to accelerate to a safe 48 mph within a 21–42 m ground roll, and thrust can never replace airspeed — at zero forward speed the wing makes no lift, so 'suspension' by the wing exists only above the stall speed.

The analysis also finds that putting the booster at the very back moves the centre of gravity about 43–60 mm behind the neutral point (static margin ≈ −32 to −45 % MAC), making the aircraft statically unstable in pitch without roughly 415 g of nose ballast. A mid-fuselage fan with a thrust tube is recommended instead.

Keywords: low Reynolds number, XFOIL, laminar separation bubble, thrust required, EDF, static margin, FDM-printed airframe

Nomenclature

Symbol	Meaning	Units
AR	Wing aspect ratio, b^2/S (= 2.55)	–

Symbol	Meaning	Units
b	Wing span (0.305)	m
$\bar{c}$	Mean aerodynamic chord, MAC (0.134)	m
CL, CD	3D lift and drag coefficients	–
Cl, Cd	2D (section) lift and drag coefficients, from XFOIL	–
CD0	Zero-lift (parasite) drag coefficient	–
e	Oswald span efficiency (0.75 assumed)	–
K	Induced drag factor, $1/(\pi\, e\, AR)$ (= 0.167)	–
L, D, W, T	Lift, drag, weight, thrust	N (or gf)
gf	Gram-force, 1 gf = 9.81 mN	–
q	Dynamic pressure, $\frac{1}{2}\rho V^2$	Pa
Re	Chord Reynolds number, $V\, c / \nu$	–
S	Reference wing area (0.03655)	m^2
V, Vs, Vmd	Airspeed, stall speed, minimum-drag speed	m/s
α	Angle of attack	deg
ρ, ν	Air density (1.225), kinematic viscosity (1.5×10^{-5})	kg/m^3, m^2/s
x_cg, x_np	Centre of gravity and neutral point, measured aft of the nose	mm
SM	Static margin, $(x_{np} - x_{cg})/\bar{c}$	% MAC

1. Introduction

Small 3D-printed jet models are attractive as test articles: the geometry is parametric, parts are cheap to replace, and a crumple nose can be swapped after each hard landing. Their aerodynamics, however, are not those of the full-size aircraft. A 1:35 scale F-35 planform flies at chord Reynolds numbers of 10^5–2×10^5, where thin sections develop laminar separation bubbles near the leading edge and lose lift far earlier than thin-airfoil theory predicts. The airframe is also heavy for its wing area (PLA shell plus electronics gives about 131 g/dm^2 of wing loading, roughly twice that of a typical foam park jet).

The question addressed here is practical: **if the only propulsion is a booster at the tail, how much thrust does it need so that the wings carry the full weight?** In steady level flight lift equals weight and thrust equals drag, so the answer is the drag curve of the aircraft at the lift coefficient its weight demands. That curve depends on the section data, which is why the wing and tail sections were analysed in XFOIL rather than taken from thin-airfoil theory.

The paper also checks two things the thrust number alone hides: the static thrust needed to reach flying speed, and the effect of a tail-mounted motor on pitch stability. This analysis supersedes the stall figures in the earlier project report, which assumed a CLmax of 0.8–1.0; XFOIL shows the realistic value for this wing is about 0.65.

2. Configuration

The airframe is the parametric Onshape model "F-35 RC Airframe" at its default parameters, printed in PLA with 1.2 mm walls (0.8 mm for the crumple nose). It splits into five parts that each fit a 256 × 256 mm bed, joined with heat-set brass inserts (M3 at the fuselage joint, M2 at the wing roots) and two 3 mm carbon spars.

Quantity	Value
Overall length	446 mm
Wing span	305 mm
Reference wing area S (trapezoid through the fuselage)	365.5 cm²
Aspect ratio	2.55
Mean aerodynamic chord	133.6 mm, leading edge at x = 189 mm
Leading-edge / trailing-edge sweep	34° / 14°
Wing section	NACA 0006 (6 % thick, symmetric), 0° incidence
Horizontal tail / canted fins	NACA 0004, 65.6 cm² exposed / 2 × 34 cm² at 25° cant
Printed parts (5)	01 nose, 02 forward fuselage, 03 aft fuselage, 04/05 wings

Quantity	Value
Estimated all-up mass	478 g (PLA parts ≈ 208 g, 50 mm EDF 77 g, 45 A ESC, 3S 1300 mAh pack, two 9 g servos, receiver, hardware)
Wing loading	131 g/dm² (12.8 N/m² × 10)
Centre of gravity (tail EDF)	x = 258 mm (51 % MAC), z = +3.8 mm

The 'booster' studied is a 50 mm EDF installed at the tail, exhausting through the nozzle, with its thrust line on the fuselage centreline. Two sensitivities on mass are carried through the paper: 420 g (lighter print, smaller battery) and 600 g (heavier electronics or ballast).

Design point. Wing loading drives everything that follows. At 131 g/dm² the aircraft is in the territory of fast sport jets, not park flyers, so the stall speed is high and the thrust needed to *reach* flying speed is far larger than the thrust needed to *stay* there.

3. Method

3.1 Section aerodynamics (XFOIL). XFOIL 6.99 (Drela, 1989) was run in viscous mode on the NACA 0006 and NACA 0004 sections with 200 panels, at Re = 40k, 60k, 100k, 150k, 200k and 300k, sweeping α from 0° to 16° in 0.25° steps. Two transition models were used:

- **Clean:** free transition, e^n method with N_crit = 9 (smooth surface, calm air).
- **Tripped (rough print):** forced transition at 10 % chord on both surfaces, representing FDM layer lines that trip the boundary layer near the leading edge.

XFOIL was run in the cloud analysis workspace (Linux build, headless) because the local Mac installation is not reachable from that environment; the code and inputs are identical. Points that failed to converge were discarded, not interpolated.

3.2 Three-dimensional lift. The 2D lift slope from XFOIL (a_0 = 6.73 /rad) was corrected for the low aspect ratio and sweep with the Helmbold/DATCOM relation, giving CLα = 3.11 /rad (0.054 /deg) for the wing. Maximum lift was taken from the XFOIL section Cl at the onset of leading-edge separation at the stall Reynolds number, reduced for sweep, giving CLmax = 0.65 (range 0.55–0.75 carried as uncertainty).

3.3 Drag build-up. Parasite drag follows Raymer's component method: CD0 = Σ Cf·FF·Q·S_wet / S. Skin friction uses the laminar (1.328/√Re) or turbulent ($0.455/(\log_{10}\mathrm{Re})^{2.58}$) flat-plate value, weighted by the XFOIL transition location; form

factors use the thickness ratio for the lifting surfaces and FF = 0.9 + 5/f^1.5 + f/400 for the fuselage (fineness ratio f). Wetted areas come from the CAD model (fuselage plus canopy 787 cm²). A 10 % allowance covers joints, nozzle base drag and insert bosses. Drag due to lift uses the parabolic polar CD = CD0 + K·CL² with e = 0.75 (K = 0.167), plus the XFOIL section profile-drag rise with Cl.

3.4 Thrust required. For steady, level, unaccelerated flight, L = W and T = D. For each airspeed V, CL = W/(qS) and T_req(V) = qS·CD(CL, Re). The minimum thrust is W/(L/D)max at the minimum-drag speed Vmd.

3.5 Thrust available, ground roll and stability. EDF thrust is modelled with momentum theory as $T(V) = T_0(1 - V/Ve)$, $Ve = \sqrt{(T_0/\rho A)}$ (Ve ≈ 62 m/s for the 50 mm unit). The ground roll integrates $m \cdot dV/dt = T(V) - D - \mu(W - L)$ with μ = 0.04 to a lift-off speed of 1.2 Vs. The stick-fixed neutral point combines the wing and tail aerodynamic centres with a downwash gradient dε/dα = 0.4–0.5 and a fuselage destabilising term (Gilruth–White, K_f = 0.012–0.020); the CG comes from the CAD part centroids plus component positions.

4. Lift and Drag Results

4.1 XFOIL section data, NACA 0006. The table summarises the XFOIL polars. 'Cl max' is the highest converged lift; for Re ⩾ 150k the solution stops converging at 5.5–6°, which is the signature of a leading-edge separation bubble bursting.

Re	Transition	Cd at α = 0°	Cl / Cd at α = 4°	Max Cl / Cd	Highest converged Cl (α)
40k	clean	0.0158	0.438 / 0.0246	19	0.83 (15.8°, gradual)
60k	clean	0.0130	0.438 / 0.0217	26	0.82 (16°, gradual)
100k	clean	0.0101	0.437 / 0.0184	33	0.65 (6.0°)
100k	tripped	0.0160	0.430 / 0.0205	22	0.82 (16°, gradual)
150k	clean	0.0083	0.436 / 0.0160	38	0.59 (5.5°)
150k	tripped	0.0142	0.430 / 0.0181	24	0.76 (13.5°)
200k	clean	0.0073	0.437 / 0.0145	40	0.61 (5.8°)
300k	clean	0.0060	0.439 / 0.0126	66	0.59 (5.8°)

Three points matter for the design. The lift slope is almost independent of Re (Cl ≈ 0.11 per degree), so XFOIL agrees with thin-airfoil theory below stall. Printing roughness roughly doubles the zero-lift section drag (0.0083 → 0.0142 at 150k). And the useful maximum lift

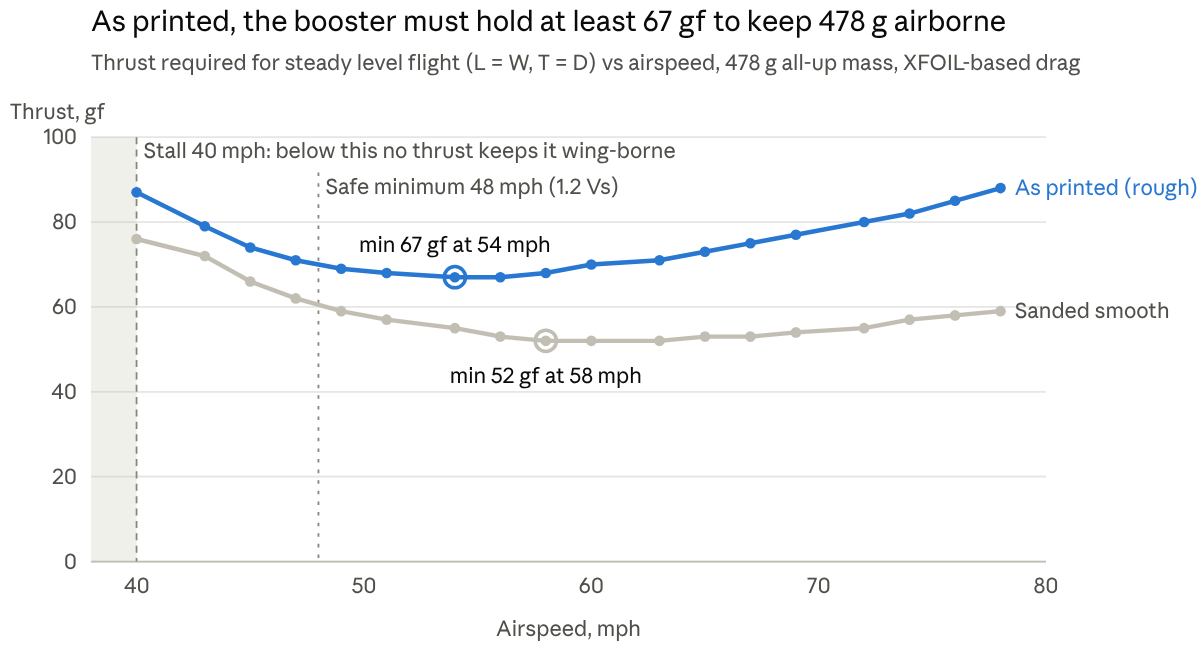
of a 6 % section is low: about 0.6–0.65 at flight Reynolds numbers, not the 1.0+ that thicker RC sections reach.

4.2 Whole-aircraft coefficients. With CLα = 0.054 /deg the wing reaches CLmax ≈ 0.65 at about 11–12° angle of attack. The component build-up gives:

Speed	Re (MAC)	CD0 clean	CD0 rough (as printed)
10 mph (4.5 m/s)	40k	0.032	0.044
45 mph (20 m/s)	178k	0.019	0.030
50 mph (22.4 m/s)	200k	0.019	0.029
67 mph (30 m/s)	267k	0.017	0.027

The fuselage contributes about 45–50 % of CD0, the wings about 30 % and the four tail surfaces about 25 %. At the 478 g design weight and 55 mph the aircraft flies at CL ≈ 0.35 and α ≈ 6.4°, with CD ≈ 0.050 and L/D ≈ 7.

5. Thrust Required to Stay Aloft



perf.py thrust-required curves from XFOIL 6.99 polars (NACA 0006, Re 100k–300k) and Raymer drag build-up · 478 g, sea-level ISA

5.1 The answer. To keep the 478 g airframe suspended by its wings in steady, level flight, the tail booster must deliver at least **67 gf (0.66 N) as printed, or 52 gf (0.51 N) if the surfaces are sanded smooth**. That minimum occurs at the minimum-drag speed, 55 mph (rough) to 60 mph (smooth), where L/D is 7.1 and 9.1 respectively. The thrust-to-weight ratio needed is only 0.11–0.14: the wing does almost all the work, and the booster only has to cancel drag.

The curve is U-shaped. To the left of the minimum, induced drag dominates: slowing toward stall forces a higher angle of attack, and at the safe minimum speed of 48 mph the requirement rises to 61–70 gf. To the right, parasite drag grows with V^2 and the requirement climbs to 59–88 gf by 78 mph.

5.2 What thrust cannot do. Lift depends on airspeed, not on thrust. Below the 40 mph stall speed the wing cannot produce 478 g of lift at any angle, so no amount of horizontal thrust from a rear booster keeps the aircraft wing-borne. Holding it up at zero airspeed would need a thrust vector pointing up and at least equal to the weight ($T \geqslant 478$ gf), which is a hover, not wing-borne flight.

5.3 Getting to flying speed. The sustained number is small, but the booster must first accelerate the aircraft from rest. With EDF thrust lapse included, the ground roll to 48 mph is:

Static thrust	Ground roll	Time to 48 mph
450 gf (T/W 0.94)	42 m	3.5 s
770 gf (T/W 1.6)	21 m	1.8 s

A hand launch is not realistic at 48 mph. A paved runway or a bungee/catapult launch is required.

5.4 Sensitivity to mass. Thrust required scales roughly with weight, and stall speed with its square root:

All-up mass	Stall speed	Min-drag speed (rough)	Min thrust, rough / smooth
420 g	37.6 mph	50 mph	60 / 46 gf
478 g	40.2 mph	55 mph	67 / 52 gf
600 g	45.0 mph	57 mph	82 / 65 gf

5.5 Recommended booster. A 50 mm EDF of about 770 gf static thrust (≈ 77 g, ≈ 38 A on 3S, with a 45 A ESC). It gives a 21 m ground roll, a large climb reserve (4.6 N available against 0.66 N required at 55 mph), and about 16 minutes of level cruise on a 1300 mAh 3S pack (jet power ≈ 19 W, electrical ≈ 43 W, ≈ 3.9 A). The cruise current is an estimate using a 45 % overall fan efficiency and should be confirmed with a wattmeter.

6. Stability and Thrust-Line Effects of a Tail Booster

6.1 Thrust line. The fan axis sits at z ≈ +4.3 mm and the CG at z = +3.8 mm, so the thrust line passes within about 0.5 mm of the CG. At 0.66 N cruise thrust the resulting pitching moment is negligible (< 0.001 N·m), and power changes will not pitch the nose up or down noticeably. This is a genuine advantage of a centreline tail booster.

6.2 Centre of gravity versus neutral point. The problem is mass, not thrust. Placing the 77 g fan at x = 415 mm, together with the ESC at 330 mm and the servos at 366 mm, puts the CG at **x = 258 mm (51 % MAC)**. The stick-fixed neutral point, from the wing and tail aerodynamic centres with downwash dε/dα = 0.4–0.5 and the fuselage destabilising term, lies at **x = 198–215 mm (7–20 % MAC)**. The CG is therefore 43–60 mm *behind* the neutral point:

Layout	CG	Static margin	Nose ballast for SM = +10 %	Flying mass
Tail EDF (as studied)	258 mm	−32 to −45 % MAC	415 g at x = 100 mm	893 g
Tail EDF, SM = +15 %	–	–	492 g	970 g
Mid-fuselage EDF + thrust tube	223 mm	−6 to −19 % MAC	225 g	723 g

A negative static margin means a nose-up disturbance grows instead of decaying; an RC pilot cannot fly it without a gyro stabiliser and it will usually depart on the first launch.

6.3 Consequences of ballasting. Ballast fixes the stability but feeds back into the thrust question. At 893 g the stall speed rises to about 55 mph, the minimum sustained thrust to about 125 gf, and the ground roll with a 770 gf fan stretches to 88 m (229 m with 450 gf). The tail booster position is therefore the single largest driver of the thrust requirement, more than the airfoil or the surface finish.

6.4 Recommended layout changes (in order of effectiveness):

1. Move the EDF forward to about x = 250–280 mm and exhaust through a thin-wall thrust tube to the nozzle. This removes about 190 g of the ballast need.
2. Move the wing aft using the Onshape parameters (the neutral point moves with it): the neutral point moves aft by most of the wing shift, while the CG moves only by the wing's share of the mass (about 10 %).
3. Put the battery as far forward as the crumple nose allows (x ≈ 90–100 mm) and print the aft fuselage in lightweight foaming PLA.
4. Add intakes. A rear EDF needs inlet area of roughly 90–100 % of the fan swept area (≈ 16–18 cm² for a 50 mm fan); the current model has none, and an EDF starved of air loses much of its static thrust.

7. Sensitivity Analysis

The minimum thrust in level flight is $T_{min} = W / (L/D)_{max} = 2W\sqrt{(C_{D0} \cdot K)}$, so it depends on weight, parasite drag and span efficiency, but **not on air density**. Density changes only the speed at which the minimum occurs.

Parameter varied	Range	Effect on minimum thrust (478 g)	Effect on stall speed
Surface finish	sanded → as printed	52 → 67 gf (+29 %)	none
Mass	420 → 600 g	60 → 82 gf (rough)	37.6 → 45.0 mph
CLmax (3D)	0.75 → 0.55	none	37.4 → 43.7 mph
Oswald efficiency e	0.85 → 0.65	63 → 72 gf	none
Air density	sea level → hot day / 1500 m (ρ = 1.06)	none	+7.5 % (≈ 43 mph)

Surface finish and mass are the two levers the builder controls. Sanding and filling the wing and tail surfaces (or printing them with a smooth top skin and ironing) saves about 15 gf of continuous thrust, which is about 20 % of the cruise power. Every 100 g saved lowers the minimum thrust by about 13 gf and the stall speed by about 4 mph.

The largest uncertainty is CLmax. XFOIL predicts a leading-edge bubble stall at Cl ≈ 0.6 for Re ≥ 150k, but the 34° swept, low-aspect-ratio planform may gain some vortex lift at high angle of attack, which would lower the real stall speed. The 40 mph figure should be treated as conservative within the 37–44 mph band.

8. Limitations and Validation Plan

8.1 Limitations.

- **2D to 3D.** XFOIL is a two-dimensional panel method with an integral boundary layer. Sweep, tip losses and the fuselage are added with empirical corrections, not computed. Fuselage lift and wing–body interference are ignored.
- **Stall prediction.** Above about 6° at Re ≥ 150k XFOIL stops converging as the leading-edge bubble bursts. The real stall may be more gradual, and low-aspect-ratio vortex lift is not modelled.
- **Roughness model.** FDM layer lines were modelled as a boundary-layer trip at 10 % chord. Real ridges also add form drag, so the 'rough' case may still be optimistic for coarse layer heights.
- **Propulsion.** The EDF thrust lapse uses ideal momentum theory. Inlet and exhaust-duct losses, which can cost 10–30 % of static thrust in a poorly ducted installation, are not included.
- **Mass and CG** are estimates from CAD volumes and catalogue component masses; the as-built aircraft must be weighed and balanced.
- XFOIL was run in the cloud analysis workspace (XFOIL 6.99, Linux build) because the local installation could not be reached from that environment. Results should match the local build; the input decks can be rerun locally to confirm.

8.2 Validation plan.

1. **Static thrust:** measure the installed fan on a kitchen-scale thrust stand with the intakes and nozzle fitted, and log current with a wattmeter.
2. **Weight and balance:** weigh the finished aircraft and find the CG by suspension or balance points; recompute the static margin before any flight.
3. **Drag:** car-top test rig (aircraft on a load-cell sting above the roof, in calm air at 40–60 mph by GPS) to measure drag and lift at fixed angles; compare against Section 4.
4. **Glide test** (only after stability is confirmed): a level hand-toss over grass checks trim, not performance.

5. **First powered flight** from a paved runway or catapult, logging GPS speed and current to verify the 55 mph cruise and the 16 min endurance estimate.

9. Conclusions

1. **Thrust to stay aloft:** a tail-mounted booster needs only **67 gf (0.66 N) as printed, or 52 gf (0.51 N) sanded smooth**, to keep the 478 g airframe suspended by its wings in steady level flight, at the minimum-drag speed of 55–60 mph. That is a thrust-to-weight ratio of 0.11–0.14.
2. **Airspeed is the price of wing-borne flight:** the NACA 0006 wing reaches only CLmax ≈ 0.65 at flight Reynolds numbers (XFOIL), so the aircraft stalls at about 40 mph. Below that speed no horizontal thrust keeps it up; only T ⩾ W (478 gf) pointed upward would.
3. **Thrust to get aloft:** 450–770 gf of static thrust is needed for a 42–21 m ground roll. A 50 mm EDF of about 770 gf is recommended; it gives a large climb reserve and about 16 min of cruise on a 1300 mAh 3S pack.
4. **The tail position is the critical risk:** it puts the CG 43–60 mm behind the neutral point (static margin ≈ −32 to −45 % MAC). Fixing it with ballast nearly doubles the mass and the thrust required. A mid-fuselage fan with a thrust tube, an aft wing shift and intakes are needed before flight.
5. **Surface finish matters:** layer-line roughness raises cruise thrust by about 29 %; sanding or ironed skins are worth the effort.
6. The stall figures in the earlier project report (30–34 mph, ≈ 19°) are superseded: XFOIL gives about 40 mph and 11–12°.

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